SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I M. Ramprasadh 192511355 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Ramprasadh

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. Even Communication Security - Caesar Cipher

a) Encrypt the message "MEET AT NOON" using shift of 4 positions.

AIM

To encrypt the given plaintext using the Caesar Cipher with a shift key of 4.

PRINCIPLE

The Caesar Cipher is a classical substitution technique in which each letter is shifted forward by a fixed number of position in the alphabet

ENCRYPTION FORMULA

$C = (P + K) \bmod 26$ where, P - plaintext
K - Shift key (4)
C - Cipher text

MEET AT NOON
12 4 4 19 0 19 13 14 14 N

$K = 4$

$P = M \Rightarrow C = (P+K) \bmod 26 = (12+4) \bmod 26$
$C = 16 = Q$

$P = E \Rightarrow C = (4+4) \bmod 26 = 8 \bmod 26$
$C = 8 = I$

$P = E \Rightarrow C = (4+4) \bmod 26 = 8 \bmod 26$
$C = 8 = I$

$P = T \Rightarrow C = (19+4) \bmod 26 = 23 \bmod 26$
$= 23 = X$

$P = A \Rightarrow C = (0+4) \bmod 26 = 4 \bmod 26$
$= E$

$P = T \quad C = (19+4) \bmod 26$
$= 23 = X$

$P = N \quad C = (13+4) \bmod 26 =$
$= 17 \bmod 26 = R$

... may encrypted as THIS IS

P=O ⇒ C = (14+4) mod 26

= 18

=S

P=O C = (14+4) mod 26

= 8

P=N C = (13+4) mod 26 = 17

=R

MEET AT NOON ⟶ QIIX EX RSSR

ain how retrieve the original message.

rmula

P = (C-k) mod 26

C=Q ⇒ P = (16-4) mod 26 =12

= M

C=I ⇒ P = (8-4) mod 26 = 4

=E

C=I ⇒ P = (8-4) mod 26 = 4

=E

C = X ⇒ P = (23-4) mod 26 = 19

= T

C = E ⇒ P = (4-4) mod 26 = 0

= A

C = X ⇒ P = (23-4) mod 26 = 19

= T

C = R ⇒ P = (17-4) mod 26 = 13

= N

C = S ⇒ P = (18-4) mod 26 = 14

C = S

=O

⇒ P = (18-4) mod 26 = 14

=O

C = R ⇒ P = (17-4) mod 26 = N

QIIX EX RSSR ⟶

MEET AT NOON

∴ Therefore the original message is successfully recovered using the correct shift key.

2. Intercepted suspicious message

a)

PRINCIPLE

The Atbash cipher is a classical substitution cipher in which each letter is replaced by its corresponding letter in the reverse alphabet. For example, A is replaced by Z, B, by Y, C by X and so on.

Given : GSRH RH Z HVXIVG

original	A	B	C	D	E	F	G	H	I	J			
cipher	Z	Y	X	W	V	U	T	S	R	Q			

GSRH → THIS

RH → IS

Z → A

HVXIVG → SECRET

GSRH RH Z HVXIVG → THIS IS A SECRET

b) Justify the answer using letter pattern

The Atbash cipher uses a fixed reverse alphabet mapping. Each cipher letter is replaced according to this rule and the resulting words form a meaning full sentence.

* The ciphertext has four words with length 4, 2, 1, and 6
* The word RH decrypts to IS, a common English word
* The letter mapping follows a consistent English when the Atbash mapping is applied
* It follows a reversed alphabet pattern

RESULT

The intercepted message was successfully decrypted as THIS IS A SECRET.

3. Secure messaging system - Vigenere cipher

Demonstrate the encryption process

AIM

TO encrypt the plaintext "HELLO" using the vigenere cipher with the keyword "KEY".

PRINCIPLE

The vigenere cipher is a polyalphabetic substitution cipher that uses a keyword to determine the shift applied to each plaintext letter

ENCryption formula

$C_i = (p_i + k_i) \bmod 26$ where,

p_i = plaintext letter value

k_i = key letter value

c_i = cipher text

HELLO
7 4 11 11 14

KEY
10 4 24

P = H C = (7+10) mod 26 = 17 mod 26
= 17 = R

P = E
C = (4+4) mod 26 = 8
= I

P = L C = (11+24) mod 26 = 9
= J

P = L C = (11+10) mod 26 = 21
= V

P = O C = (14+4) mod 26 = 18
= S

HELLO ⟶ RIJVS

b) How the repeating key word works

* The keyword KEY contains three letter, but the plain text contains five letters. Therefore, the key repeats as KEYKE.

* Each key letter determine the shift applied to its corresponding plain text letter.

For example:

H+K = R
E+E = I
L+Y = J
L+K = V
O+E = S

Conclusion:

The vigenere cipher uses different shifts determined by the keyword making it more complex than a basic Ceaser cipher. However it is not secure enough for modern confidential communication.

4. Suspicious Digital Asset Investigation - steganography

AIM

To identify whether a company logo contains hidden confidential information.

Principle:

Steganography is the technique of hiding secret information inside an ordinary file, such an image, audio file or video. The image may appear normal even when hidden data is present.

Step 1: Investigators inspect pixel examine the image's file size, format, metadata and file structure to identify unusual information

Step 2: Investigation inspect pixel patterns and least significant bits, which can be modified to store secret information within an image

Step 3:

Tools such as ExifTools an examine image metadata, while Binwalk can help identify embedded or append data.

Step 4: Investigators analyze pixel distribution and unusual patterns that may indicate hidden information.

RESULT :

The finding should be verified against the original file or a trusted copy because unusual characteristics do not automatically prove that a file contains secret data

b) TWO techniques to strengthen security

1. Encrypt the secret data before embedding it.

The confidential message could be encrypted using a strong modern encryption algorithm, such as AES, before being embedded in the image. They should not be able to read it without the decryption key.

2. Use robust Steganographic method.

Choose a method appropriate to the image format that minimize detectable statistical changes. The method should be evaluated against known steganalysis techniques.

RESULT

Combining encryption and decryption steganography improve communication security by protecting the message content and concealing their presence.

5. Legacy Encryption - Three Rail fence cipher

a) Determine the original message

WECR I TEERSOEE FAOCAIVDEN

This matches three level Rail fence cipher.

Principle

In a rail fence cipher, plain text letters are written in a zigzag pattern across fixed number of rails.

Step 1 : Identify no. of rails

Number of Rails = 3

Step 2 : Reverse the zig zag

Rail 1 : 7 letters
Rail 2 : 12 letters
Rail 3 : 6 letter

Rail 1 · W E · R I T E

Rail 2 E R D S O E E F A O C ·

Rail 3 A I V D E N

Step 3: Reconstruct the zig zag.

Place the extracted letters back into the three rail zig zag position and read them diagonally in the original order.

WECRITEERSOEEFAOCAIVDEN → WE ARE DISCOVERED FLE AT ONCE

b) why is this method vulnerable in modern communication.

1. NO substitution
2. preserves letter frequencies
3. Small key spaces
4. known plaintext attacks
5. NO authentication
6. Limited confidentiality

Analysis

the Rail Fence cipher is useful for understanding transposition based encryption.

c) one parallel - enhancement

Replace legacy transposition cipher with AES-256 authenticated encryption, such as AES-GCM.

This provides confidentiality and integrity protection, helping prevent unauthorized reading and undetected modification of messages.